



Sony's new AAA mobile game

Sony's new PlayStation Studios Mobile Division is working on a AAA title for phones. <https://digit.in/sep22-41>



TSMC to start 3nm chip production

TSMC will soon produce an ultra-advanced 3-nanometer (nm) chip amid supply-chain woes. <https://digit.in/sep22-42>

AMD Ryzen 3 4100 Lightweight

The other new Ryzen "4000" processor from AMD, the AMD Ryzen 3 4100 is an entry-level processor aimed at folks going for a budget build. It's still got multithreading in the entry-level, much like previous Ryzen processors and even the new hybrid processors from the competition, however, it's position in the SKU stack is a little odd considering that the budget segment still has the Athlon 3000G, a few OEM Ryzen 4000 processors and other units which are competing in a very narrow price bracket. Let's see if the Ryzen 3 4100 can make a name for itself.

SPECIFICATIONS

If we are to compare the Ryzen 3 4100 against the previous gen Ryzen 3 3100, then both processors are matched in terms of core counts and number of threads. As for the clock speeds, there's an improvement of 200 MHz in the base clock and 100 MHz in the boost clock speeds. Then there are the cache sizes which remain unchanged for L1 and L2 cache. That being said, the L3 cache has taken a massive reduction from 16 MB down to 4 MB. The Ryzen 3 3100 was a 'Zen' processor and the Ryzen 3 4100 is a 'Zen 2' processor so there's the improved architecture lending to increased performance but the drop in L3 cache is going to bite in some place or the other.

Also, both processors are built on the same 7nm process node, so there aren't exactly a lot of power efficiency metrics to talk about. Even the PCIe revision is different between the two processors and the 3100 has the better deal with PCIe 4.0 support. Lastly, the Ryzen 3 4100 does not feature a graphics die much like the 3100. If we were to personify the Ryzen 3 4100, then this processor is very content in being a budget processor and has very little ambition to do better.



PERFORMANCE

The Ryzen 3 4100 is an entry-level processor so the type of tests and the performance profiles used to benchmark are in tune with the expectations of an entry-level processor. We kick off the benchmarks with Cinebench R20 and the Ryzen 3 4100 scores around 456 points which is very much in line with the other Zen 2 processors. Zen 3 processors typically have a single core score around 600 points. Multi-core scores are around 2104 points which is also a decent score on its own. To compare against the 4100, we have the previous gen Ryzen 3 3100 and the Intel Core i3-12100F. Both processors are priced about the same. The Intel processor is also a 4-Core/8-Thread processor with slightly higher clock speeds and that does help with the Cinebench scores as the 12100F comes ahead with a clear 900+ point lead. That's about a 30 per cent higher score. The Ryzen 3 3100 also scores around 2270 points which makes it more than the 4100. This could be one of the fallouts of the reduced L3 cache.

Switching to compression and decompression benchmarks, the 7-Zip test shows a performance difference

of close to 30 per cent again when compared to the Intel 12100F with the 12100F taking the lead.

In day-to-day productivity tasks, the performance difference between the Ryzen 3 4100 and the Intel Core i3-12100F narrows down but Intel still has the lead in this scenario. While the benchmark scores are quite different, during actual use, the processor doesn't appear to be all that different in performance. Tasks such as writing, browsing and consuming content seem to be running without any hiccups.

As for gaming, the Ryzen 3 4100, like most entry-level processors, does bottleneck a high-end GPU. Then again, folks buying an entry-level processor don't pair them with a high-end GPU. Compared to the 12100F, there is a

marked difference in performance with certain eSports titles really widening that gap. CS:GO, for example, had a performance difference of close to 150 FPS which is nearly a 44 per cent difference. However, in games which are not that optimised such as Metro

Exodus, the performance difference comes down to 20 FPS. That's barely 13 per cent.

VERDICT

While on paper, the Ryzen 3 4100 might appear to be a worthy successor to the Ryzen 3 3100, the reduced L3 Cache seems to be affecting the performance quite a bit. But when you consider the fact that entry-level processors are mainly used in office machines and basic rigs, then the 4100 ends up being a decent processor. However, the Core i3-12100F is a much more value for money offering considering how much more performance it provides within the same price bracket.

—Mithun Mohandas

SPECIFICATIONS

CORES: 4 | THREADS: 8 | GPU CORES: N/A | BASE CLOCK: 3.8 GHz | BOOST CLOCK: 4.0 GHz | PROCESSOR NODE: 7nm | L3 CACHE: 4 MB | L2 CACHE: 512 KB per core | TDP: 65 W

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